Data Analytics Insights Report

This report explains the main findings from the four tasks. We have used SQL queries, Excel and Power BI to study the traffic patterns. The main aim of this project is to help the smart-city mobility team understand the busy periods and make better planning decisions.

The dataset contains **48204** records and **40575** different recorded hours. Some hours appear more than once. I have used all records, while the total hours card counts each recorded hour once.

Task 1 Yearly traffic and holidays

Recorded traffic totals fell in **2014** and **2015**, then increased in **2016** and **2017**. The largest decrease was **44.17%** between **2013** and **2014**. The highest recorded total was in **2017**, at about **35.43** million. This was **20.12%** higher than **2016**.

However, all the years do not have the same amount of data. For example, **2012** starts in October, and there are missing periods in other years. Repeated hours also affect the totals. The team should check these gaps before using the results to say that actual traffic increased or decreased.

The holiday results did not show a clear link between temperature and traffic. The record labelled New Year's Day was **4.68C** warmer in **2017** than in **2016**, but traffic fell from **1513** to **798**. The **2017** record is dated **2** January, while the **2016** record is dated **1** January. Labor Day also showed no consistent pattern. These results only cover midnight records, so they cannot describe the whole holiday. No New Year's Day record was availble for **2015**.

Task 2 Traffic statistics and temperature

The average traffic volume was **3259.82**, and the median was **3380**. The population standard deviation was **1986.84**, and the population variance was **3947533.43**. Traffic values ranged from **0** to **7280**, giving us a range of **7280**.

These results show that traffic changes a lot between records. The average alone does not show how busy or quiet a particular hour can b e. The team should study these periods seperately.

The correlation between temperature and traffic was **0.1303**. This shows a weak positive link. Warmer temperatures sometimes came with more traffic, but the relationship was small. This does not prove that temperature causes traffic to increase. Time of day and people's travel habits might also explain the pattern.

Task 3 Probability and congestion

Congestion was defined as traffic volume above **5500**. Using this rule:

- **14.73%** of records had congestion.
- **27.78%** had clear weather.
- **3.66%** had both congestion and clear weather.
- Among congestion records, **24.83%** had clear wether and **26.30%** had temperatures above **292K**.

The independence calculations did not give equal results. This indicates that the events were not exactly independent in this dataset.

The odds ratio for congestion in clear weather compared with cloudy weather was **0.7354**. Congestion had lower odds in clear weather. However, this does not prove that weather caused the difference.

Traffic volume aso does not tell us how slowly vehicles were moving.

Task 4 Power BI dashboard

The original data had **9** columns and no blank or null values. We checked the data types and added Hour, Temperature in Celsius, Traffic Category and Datecolumns.

The daily chart shows frequent changes in traffic. In **2017**, the highest hourly average was at **4** PM, at about **5798**. The lowest was at **3** am, around **382**.

Clouds had the highest average traffic at **3618.45**. Squall had the lowest at **2061.75**. The difference was **1556.70**.

The scatter plot showed no clear temperature–traffic pattern. We havce found **10** invalid(outliers)temperature readings at **−273.15°C** and hid them in the second plot.

The cards showed **40575** recorded hours, average traffic of **3259.82** and average temperature of **8.06C**. The temperature average still includes the invalid readings. All three filters worked correctly in our combined test.

Summary

The team should focus monitoring on busy afternoon hours and consider quieter hours for road maintenance. It should also check missing hours, , invalid temperatures , repeated records and zero traffic values before further analysis. Weather might help with planning, but it should be considered along with time of day and other travel information.